

Module 3: Data Organization and Manipulation using Collections

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In module 3,

- Data Handling Strategies Using Lists, Tuples, Dictionary and Set
- Data Comprehension
- Data Modification
- Grouping and Categorization
- Iterators
- Data Selection - Ordered, Unordered and Unique Data
- Searching Techniques: Linear Search, Binary Search
- Sorting Techniques: Bubble Sort, Selection Sort- Insertion Sort- Quick Sort- Merge Sort.

What is List Comprehension?

- A short, elegant way to create lists in Python.
- Instead of using loops and `append()`, we can build lists in a single line.

Example (Syntax)

```
[expression for item in iterable if condition]
```

expression → can be a function, operation, or just the item.

item → Variable representing each element in the iterable.

iterable → A sequence (list, tuple, string, range, etc.).

condition (optional) → Filter elements.

Create a list of numbers

Example (Syntax)

```
numbers = [x for x in range(10)] print(numbers)
```

Squares of numbers

Example (Syntax)

```
squares = [x**2 for x in range(10)]  
print(squares)
```

With condition (only even numbers)

Example (Syntax)

```
evens = [x for x in range(10) if x % 2 == 0] print(evens)
```

Get characters from a string

Example (Syntax)

```
chars = [ch for ch in "Python"] print(chars)
```

Uppercase only vowels

Example (Syntax)

```
vowels = [ch.upper() for ch in "education" if ch in  
"aeiou"] print(vowels)
```


Nested list - Flatten a matrix

Example (Syntax)

```
matrix = [[1, 2], [3, 4], [5, 6]] flat = [num for row in  
matrix for num in row] print(flat)
```

Nested list - Multiplication table

Example (Syntax)

```
table = [[i*j for j in range(1, 6)] for i in range(1, 6)]  
print(table)
```

If-Else - Replace odd numbers with "odd"

Example (Syntax)

```
result = [x if x % 2 == 0 else "odd" for x in range(5)]  
print(result)
```

If-Else - Categorize marks

Example (Syntax)

```
marks = [90, 45, 67, 32, 88] grade = ["Pass" if m >= 50  
else "Fail" for m in marks] print(grade)
```

List of Tuples

Example (Syntax)

```
pairs = [(x, y) for x in range(3) for y in range(3)]  
print(pairs)
```

Dictionary Comprehension

Example (Syntax)

```
squares = {x: x**2 for x in range(5)} print(squares)
```

Set Comprehension

Example (Syntax)

```
unique_letters = chforchin" programming" print(unique_letters)
```

Questions

- Create a list of 5 fruits and print the second and last fruit using indexing.
- Write a program to add "Python" at the end of a list of languages.
- Given a list [10, 20, 30, 40, 50], access elements using negative indexing.
- Write a program to find the length of a list without using len() (hint: loop).
- Create a nested list of 2D matrix [[1,2],[3,4],[5,6]] and access 4.

Questions

- Write a program to find the maximum and minimum element in a list without using `max()` or `min()`.
- Reverse a list without using `.reverse()` or slicing.
- Count how many times "apple" appears in a list.
- Given two lists, merge them into one without using `+` or `.extend()`.
- Remove duplicates from a list while preserving order.

Questions

- Generate a list of squares of numbers from 1 to 10 using list comprehension.
- From a list of numbers, create a new list containing only even numbers.
- Write a list comprehension to extract the first letter of each word from: ["apple", "banana", "cherry"] $\rightarrow$ ['a', 'b', 'c'].
- Generate a list of numbers divisible by both 3 and 5 in the range 1–50.
- Create a list of (number, square) pairs for numbers 1–5.

Questions

- Flatten a nested list `[[1,2],[3,4],[5,6]]` into `[1,2,3,4,5,6]` using list comprehension.
- Write a list comprehension to extract only palindromes from: `["madam", "apple", "level", "banana"]`.
- Use list comprehension to generate a list of prime numbers from 1 to 50.
- Given a sentence "Python is fun and powerful", create a list of word lengths.
- Create a multiplication table of 5 (from 1 to 10) using list comprehension.

Questions

- From a list of dictionaries: `students = [ "name": "Geetha", "marks": 85, "name": "Ram", "marks": 67, "name": "Charan", "marks": 92 ]`
Extract the names of students who scored more than 80 using list comprehension.
- Given a list of temperatures in Celsius `[0, 20, 37, 100]`, convert them to Fahrenheit using list comprehension.
- From a paragraph, extract all unique words (case-insensitive) using list comprehension + `set()`.
- Write a program using list comprehension that generates all possible pairs of `(x,y)` where `x` is from `[1,2,3]` and `y` is from `[4,5,6]`.
- Generate Pascal's Triangle rows using nested list comprehensions